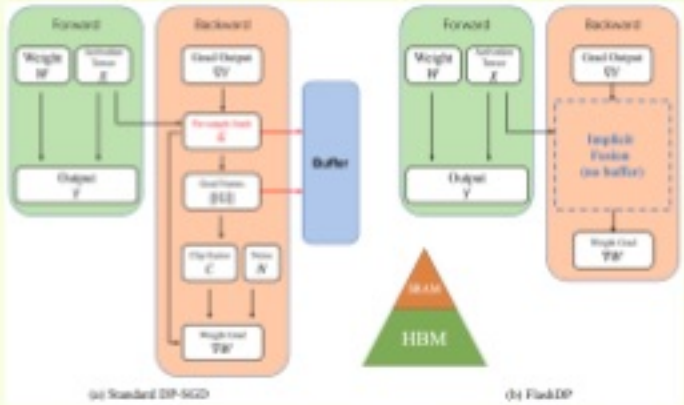
Private Training Large-scale Models with Efficient DP-SGD

Liangyu Wang, Junxiao Wang, Jie Ren, Zihang Xiang, David Keyes, Di Wang

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Motivation: DP-SGD / FlashDP



1. Challenge: Explicit methods (e.g., Opacus) allocate large HBM buffers to store per-sample gradients, breaking kernel fusion and causing frequent SRAM–HBM transfers.
2. Goal: Design a cache-friendly pipeline that minimizes HBM access and fuses gradient computation and clipping into a single kernel.

Baseline: DP-SGD

Algorithm 3 Per-Layer Gradient Processing in DP-SGD

Require: $\mathcal{L}(\theta^{(l)}, x_i)$: Loss function for layer parameters $\theta^{(l)}$ and input x_i
Require: $C^{(l)}$: Clipping threshold for layer l
Require: $\sigma^{(l)}$: Noise scale for layer l
Require: L: Total number of layers

1: **for** each layer $l = 1$ to L **do**
2: **for** $i = 1$ to B **do**
3: Compute gradient for layer l: $g_i^{(l)} = \nabla \mathcal{L}(\theta^{(l)}, x_i)$
4: Clip gradient for layer l: $\bar{g}_i^{(l)} = g_i^{(l)} \min\left(1, \frac{C^{(l)}}{\|g_i^{(l)}\|_2}\right)$
5: **end for**
6: Aggregate clipped gradients for layer l and add Gaussian noise: $\tilde{g}^{(l)} = \frac{1}{B}\sum_{i=1}^{B} \bar{g}_i^{(l)} + \mathcal{N}(0, (\sigma^{(l)} C^{(l)})^2 I)$
7: **end for**

Our Method: FlashDP

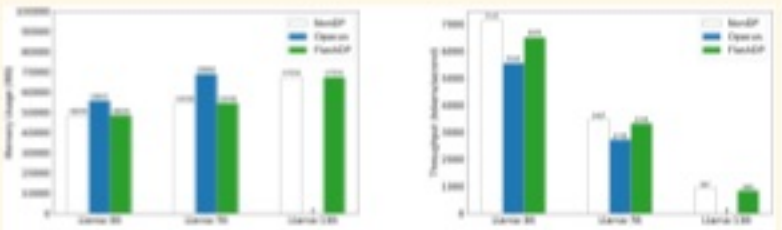


1. Unified Kernel Fusion: Computes per-sample gradients, norms, clipping, and noise addition entirely in SRAM without intermediate buffers.
2. Block-wise All-Reduce: Hierarchical reduction aggregates norms on-chip, reducing inter-block communication and global memory traffic.

Liangyu Wang
- PhD Candidate, King Abdullah University of Science and Technology (KAUST)
- Expected Graduation: 2026/2027
- Currently seeking Summer Internship or Research Scientist opportunities.

Linkedin

Experiment Results



(a) Memory Usage (b) Throughput

Model	B	Memory Usage (MB x1e4)					Throughput (tokens/sec x1e4)				
		NonDP	Opacus	GhostClip	BK	FlashDP	NonDP	Opacus	GhostClip	BK	FlashDP
GPT2-small	1	0.50	0.75(x1.50)	**0.46(x0.92)**	0.53(x1.06)	0.50(x1.00)	2.84	0.91(x0.32)	0.57(x0.20)	1.56(x0.54)	**1.83(x0.64)**
GPT2-medium	1	1.26	1.53(x1.21)	**1.12(x0.89)**	1.68(x1.33)	1.26(x1.00)	1.10	0.42(x0.38)	0.39(x0.35)	0.75(x0.68)	**0.86(x0.78)**
GPT2-large	1	2.48	3.99(x1.61)	**2.17(x0.88)**	2.73(x1.18)	2.48(x1.00)	0.58	0.25(x0.43)	0.27(x0.46)	0.40(x0.69)	**0.51(x0.89)**
GPT2-small	2	0.87	1.30(x1.49)	**0.79(x0.91)**	1.01(x1.16)	0.87(x1.00)	3.22	1.68(x0.52)	0.92(x0.29)	1.91(x0.59)	**2.32(x0.72)**
GPT2-medium	2	2.07	2.89(x1.39)	**1.87(x0.90)**	2.44(x1.18)	2.07(x1.00)	1.28	0.74(x0.58)	0.59(x0.46)	0.81(x0.63)	**1.02(x0.80)**
GPT2-large	2	3.91	4.79(x1.23)	**3.53(x0.90)**	4.81(x1.23)	3.91(x1.00)	0.68	0.38(x0.56)	0.38(x0.56)	0.45(x0.66)	**0.59(x0.87)**
GPT2-small	4	1.53	2.07(x1.35)	**1.44(x0.94)**	1.68(x1.09)	1.53(x1.00)	3.60	2.42(x0.67)	1.42(x0.39)	2.24(x0.62)	**2.59(x0.72)**
GPT2-medium	4	3.58	4.26(x1.19)	**3.33(x0.93)**	4.00(x1.12)	3.58(x1.00)	1.42	0.90(x0.63)	0.81(x0.57)	0.95(x0.67)	**1.13(x0.80)**
GPT2-large	4	6.60	-	**6.15(x0.93)**	6.60(x1.00)	6.60(x1.00)	0.76	-	0.50(x0.66)	0.53(x0.70)	**0.64(x0.84)**
GPT2-small	8	2.86	3.44(x1.20)	**2.72(x0.95)**	2.86(x1.00)	2.86(x1.00)	3.80	2.64(x0.69)	1.92(x0.51)	2.40(x0.63)	**2.72(x0.72)**
GPT2-medium	8	6.60	-	**6.24(x0.95)**	6.60(x1.00)	6.60(x1.00)	1.52	-	0.99(x0.65)	1.03(x0.68)	**1.19(x0.78)**
GPT2-large	8	-	-	-	-	-	-	-	-	-	-